

NILAKSHAN KUNANANTHASEELAN

Research Scientist | Multimodal AI, Computer Vision & Generative Models

nilakjhc@gmail.com | Melbourne, Australia | Full Australian work rights

nilakshankunananthaseelan.github.io | linkedin.com/in/knilakshan20 | github.com/NilakshanKunananthaseelan

Profile

Research scientist with 7+ years across machine learning engineering and applied research, specialising in computer vision, deep learning, LLMs, and VLMs. PhD contributions span model adaptation, machine unlearning, behavioural control, alignment correction, and construction-domain vision, including an AAAI 2026 oral paper and AAAI 2024 publication. Combines first-principles investigation with end-to-end development of models, multi-agent systems, evaluation, and deployment. Current research interests include multimodal reasoning, continual learning, generative models, and multimodal agents.

Research Areas & Technical Expertise

Research Areas: multimodal reasoning, VLMs/MLLMs, generative models, vision agents, machine unlearning, model adaptation

Computer Vision: deep learning, object detection, image segmentation, visual recognition, person re-identification, SAM/SAM2, CLIP

ML Development & Lifecycle: data preparation, pseudo-labelling, training and evaluation, hyperparameter optimisation, model adaptation, integration, experiment design

Inference & Deployment: hardware-aware optimisation, accelerator inference, quantisation, pruning, compression, integer-only models, Docker, Ray, Vertex AI

Programming & Frameworks: Python, PyTorch, TensorFlow, Hugging Face, LangGraph, Flask, C++

PhD Research Contributions

PhD Candidate | Monash University, Australia

Jan 2023 - Jun 2026

Thesis: Multimodal Adaptation of Foundation Models

- **Retain-free machine unlearning:** Developed DIET, a hyperbolic-geometry method that removes targeted knowledge from vision-language models without retained training data; accepted as an oral paper at AAAI 2026. Investigated extensions to MLLMs and generative architectures.
- **Post-unlearning model alignment:** Designed a Bézier-curve optimisation method to realign model representations after unlearning while preserving targeted forgetting. Manuscript in preparation.
- **Behavioural control:** Investigated Busemann-guided activation steering as an alternative to parameter optimisation, including steering-vector projections and activation-based gating to separate forget and retain knowledge.
- **Construction-domain vision:** Co-developed PromSeg-Waste, a prompt-guided computer-vision pipeline for segmenting construction and demolition waste in cluttered imagery; bounding-box, point, and text prompts reduced annotation effort and improved performance by 9% on average over state-of-the-art baselines.
- **Retrieval-augmented adaptation:** Developed a test-time fine-tuning framework that adapts the SAM2 segmentation decoder to construction-waste and medical-imaging domains using information-guided support-sample retrieval and point-prompt pseudo-labels.
- **Language-grounded visual prompting:** Developed LaViP, a model-reprogramming method that uses language to adapt CLIP efficiently to out-of-distribution tasks; published at AAAI 2024.

Selected Applied Research Experience

Senior ML Research Engineer | Sagence AI, USA

Mar 2019 - Nov 2022

Computer vision and hardware-aware deep learning / Remote

- Developed and evaluated ResNet, FCN, YOLOv5, MIPNet, and Person-ReID models under analog-accelerator precision and noise constraints.
- Designed hardware-aware optimisation methods spanning quantisation, pruning, compression, noise compensation, and integer-only inference to preserve model robustness across accelerator conditions.
- Built a model-structure prediction tool that translated neural-network configurations into corresponding hardware-model definitions, automating model-to-accelerator integration.

Machine Learning Engineering & Applied Research

Applied Research Scientist | OpenStreamAI, Australia
Agentic AI and vision-language systems

Mar 2025 - Present

- Developed a semantic-first synthetic-data generation pipeline for fine-tuning document-understanding VLMs on enterprise forms; research manuscript under review.
- Collaborated with the team to research, design, and build a multi-agent system from scratch for complex decision-making and workflow automation, translating client requirements into system architecture and working functionality.
- Designed and evaluated an LLM-based verification component to assess system outputs and improve workflow reliability.
- Evaluated smaller open-source models and custom agentic architectures to balance task performance, reliability, and deployment cost.
- Developing an agent-evaluation pipeline covering task completion, output reliability, and regression testing to support adaptation to evolving user requirements.

Team Manager, ML Research Team | ExentAI, UK
Applied LLM systems and rapid product development / Remote

Jun 2024 - Dec 2024

- Led development of an LLM-powered copilot for real-estate document analysis, reducing manual data-processing time by 40%, and translated client requirements into iteratively evaluated MVPs.
- Designed automated contract and invoice processing workflows that improved information-retrieval efficiency by 35% and reduced manual data entry.
- Developed an agentic RAG system using a private LLM and custom document parser, achieving 92% response accuracy on the project evaluation set.
- Built an OCR pipeline for scanned Indic-language documents and multi-agent systems using small language models for sentiment analysis, named-entity recognition, and keyword extraction.

Research Intern | Northwestern University, USA
Computational and Integrative Pathology Group / Remote

Feb 2021 - Jun 2023

- Prepared and curated breast-cancer datasets, applying quality and integrity checks before model development.
- Developed multi-task survival models with domain-adversarial training to improve generalisation across data distributions.
- Built a Ray-based hyperparameter-optimisation package for repeatable model tuning and evaluation.

Undergraduate Research Intern | CooperLab, Emory University, USA
Computer vision for digital pathology / Remote

Feb 2018 - Oct 2019

- Developed and evaluated a two-stage region detector and classifier for white-blood-cell counting using more than 10,000 annotated cells, achieving 0.959 precision-recall AUC for detection and 0.982 ROC-AUC for classification on non-neoplastic samples.
- Integrated the resulting model into HistomicsTK to support automated cell counting in digital-pathology workflows.

Education

PhD, Electrical and Computer Systems Engineering
Monash University, Australia | Thesis submitted; examination pending

2023 - 2026

Bachelor of Science, Electronic and Telecommunication Engineering
University of Moratuwa, Sri Lanka | Dean's Honour List | GPA: 3.8

2014 - 2018

Selected Publications & Recognition

- **DIET: Machine Unlearning on a Data-Diet.** AAAI Conference on Artificial Intelligence, 2026. *Oral paper.*
- **LaViP: Language-Grounded Visual Prompting.** AAAI Conference on Artificial Intelligence, 2024.
- Awarded A\$33K in computing resources by Monash eResearch and DUG for *Personalizing Foundational Models via Multimodal Signals*.
- **Optimizing Waste Handling with Interactive AI: Prompt-Guided Segmentation of Construction and Demolition Waste Using Computer Vision.** *Waste Management*, 2024.
- AAAI Student Scholar Program recipient, 2024 and 2026.